

Auteur: Sabawoon Enayat
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 25.15

Opgave: $\int \frac{2x^2 - x + 1}{2x + 1} dx$

$$\begin{array}{r|l} \begin{array}{rrr} 2x^2 & -x & +1 \\ -2x^2 & -x & \\ \hline & -2x & +1 \\ & 2x & +1 \\ \hline & & 2 \end{array} & \begin{array}{rr} 2x & +1 \\ x & -1 \end{array} \end{array}$$

$$= \int x - 1 + \frac{2}{2x + 1} dx = \int x dx - \int dx + 2 \int \frac{dx}{2x + 1}$$

$$\Rightarrow t = 2x + 1, t' = 2 \Rightarrow dx = \frac{dt}{2}$$

$$= \frac{x^2}{2} - x + \int \frac{dt}{t} = \frac{x^2}{2} - x + \ln |t| + c$$

$$= \frac{x^2}{2} - x + \ln |2x + 1| + c$$

References